5 Results

Extracted from: Adaptive Hop and Batch Control and Parallel Ensembles for Tree Parity Machine Key Exchange: A Security and Efficiency Study (A. Kumari et al.)

5.1 Overview of the eight exchanges

Table 3 lists the eight exchanges. All eight synchronize before the round limit, derive the same key at Alice and Bob, and are reported as attack failed. The simulator shows one communication round more than the learning rounds (40,762 against 40,761 for R1), which we read as the final reconciliation and confirmation round. The wire bits equal twice the tau values and the updates equal the sum of the step usage counts in every run, so the summaries are internally consistent.

Run	Rounds	Tau values	Updates	Wire bits	Equal before (%)	Final (%)	Peak ▲ (%)
R1	40,761	40,761	32,630	81,522	100.00	2.9	36
R2	5,655	56,550	44,311	113,100	100.00	5.3	14
R3	47,378	47,378	38,647	94,756	100.00	4.1	14
R4	4,497	44,970	36,180	89,940	99.97	6.1	14
R5	78,290	166,729	133,033	333,458	100.00	2.3	15
R6	7,261	154,280	122,821	308,560	100.00	2.3	5
R7	12,949	67,235	60,884	134,470	99.90	2.9	22
R8	13,366	73,930	69,430	147,860	99.97	1.5	21

Table 3. Results of the eight exchanges. Rounds are batches (the slowest pair for $P = 3$). Equal weights after reconciliation are 100.00% in all runs. Final is the best attacker agreement reported at the end (for R5 and R6 the weakest of the three pairs, since the attacker needs every pair). The peak marked ▲ is read by eye from the plotted curve (about ±2 points) and is not reported by the simulator.

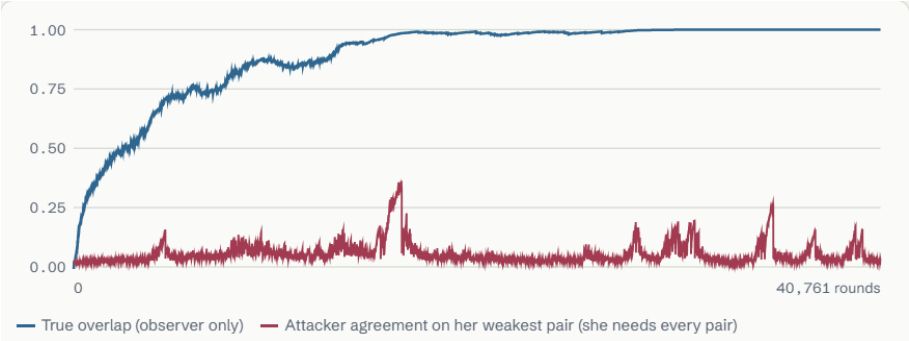


Fig. 3. True overlap of Alice and Bob (blue) and best majority-attacker agreement (red) for R1, the unbatched 1-h scheme with SCRN inputs (40,761 rounds). The attacker agreement peaks near 0.36 and ends at 2.9%.

5.2 Cost of the 1-*h* scheme: batch size and input generator (R1 to R4)

Batching trades round trips for tau values. With SCRN inputs, $B = 10$ (R2) needs 7.2 times fewer rounds than $B = 1$ (R1) but 1.39 times more tau values (56,550 against 40,761), and the share of taus that trigger an update falls from 80.1% to 78.4% refer Table 4. With fresh inputs (R3 to R4) the rounds fall 10.5 times and the tau values change by a factor of 0.95. The bandwidth penalty of batching is therefore not constant, and the input generator shows the same inconsistency: at $B = 1$ the SCRN run needs 14% fewer tau values than the fresh-input run and at $B = 10$ it needs 26% more. One seed cannot separate a generator effect from run-to-run variation, so we draw no conclusion about the generator. Figure 3 shows R1. In all four single-pair 1-h runs the true overlap settles at 1 and the best attacker agreement stays far below it.

5.3 Adaptive controller (R7, R8)

The adaptive runs are R7 (base batch 1) and R8 (base batch 10). In the 1-*h* runs the hops 1 and 3 each account for about half of the updates refer Table 4. In the adaptive runs hop 1 accounts for 74.3% and 85.6% of the updates, hop 2 for 17.9% and 8.4%, and hop 3 for 7.8% and 6.0%, so the mean step falls to 1.34 and 1.20 and the share of taus that trigger an update rises to 90.6% and 93.9%. In R8 the minimum batch size 5 (the Late setting) is used in 12,080 of 13,366 batches (90.4%), size 10 in 1,219 and size 20 in 67. In R7 only the sizes 5 and 20 occur (12,783 and 166 of 12,949 batches), the raised base size of Section 3.3. The controller changed the hop 628 and 234 times and the batch size 81 and 220 times in the two runs. The plot of R8 in Fig. 4 shows many one-batch excursions after the controller has settled, so a hysteresis of 2 updates does not remove switching.

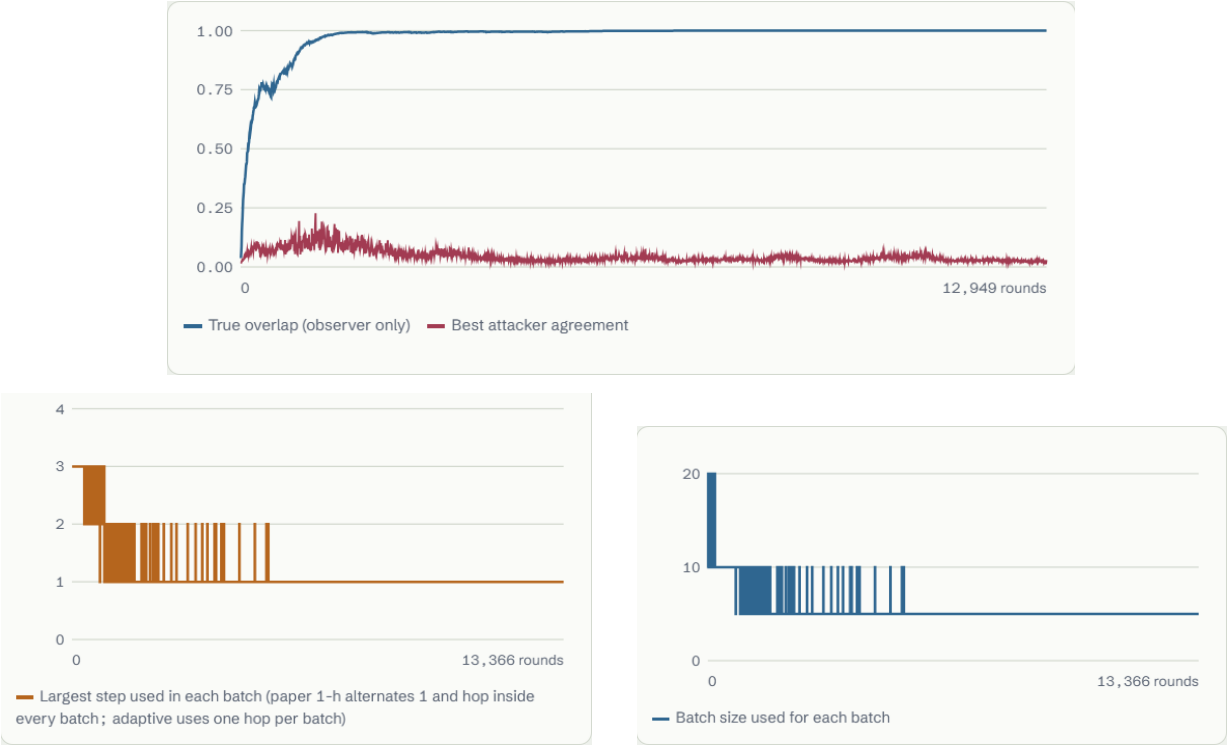


Fig. 4. Adaptive runs. Top: true overlap and best attacker agreement for R7 (base batch size 1, 12,949 rounds). Bottom: largest step and batch size used in each batch of R8 (base batch size 10, 13,366 rounds).

Run	Updates per tau (%)	Hop 1 (%)	Hop 2 (%)	Hop 3 (%)	Mean step
R1	80.1	50.4	0.0	49.6	1.99
R2	78.4	50.0	0.0	50.0	2.00
R3	81.6	50.0	0.0	50.0	2.00
R4	80.5	50.1	0.0	49.9	2.00
R5	79.8	50.1	0.0	49.9	2.00
R6	79.6	50.0	0.0	50.0	2.00
R7	90.6	74.3	17.9	7.8	1.34
R8	93.9	85.6	8.4	6.0	1.20

Table 4. Update fraction and step usage. Updates per tau is the share of taus that matched. Hop shares are shares of the updates. Mean step is the average hop over all updates.

At the start of a run the controller has the least information. In the trace table of R8 the smoothed match rate stays between 0.46 and 0.63 for the first 160 batches, the level of two unrelated machines, the hop stays at 3 and the phase alternates between Early and Moderate. At round 85 the smoothed rate of 0.49 lies below 0.50 but the phase is still Moderate, because a

change needs the same target phase on two consecutive updates. R7 behaves the same way. The smoothed rate then climbs above 0.90 and the controller moves to the Late setting.

Against the unbatched baseline (R7 against R1) the controller needs 3.1 times fewer rounds and 1.65 times more tau values and wire bits. Against the batched baseline (R8 against R2) it needs 2.4 times more rounds and 1.31 times more tau values. It is thus faster than the 1-h scheme only when that scheme is unbatched. It always takes smaller steps.

5.4 Attacker observations

No attacker came near the 98% criterion (Table 3). The highest peak is about 36% in R1, and the final values range from 1.5% to 6.1%. These values cannot rank the schemes. The agreement fluctuates within a run, so the final value is one sample of a moving curve: R1 ends lower than R2 (2.9% against 5.3%) although it has the larger peak, and the fresh-input runs end higher than their SCRN counterparts. The adaptive runs show a bump of 21% to 22% early, when the hop is 3 and the batch size is large, but R1 has a higher excursion later. We therefore do not claim that the controller lowers or raises the attacker success. We only note that its mean step is smaller, which the literature [4,11] associates with a lower attacker success.

5.5 Parallel ensemble (R5, R6)

The ensemble ends with the slowest pair refer Table 5, so its rounds are 1.92 and 1.28 times those of the single-pair runs R1 and R2, and 47% of the serial cost in both runs. The rounds are therefore not close to those of one pair here: the slowest pair needs 1.84 and 1.85 times the rounds of the fastest, and the ensemble inherits the upper tail of three synchronization times. Pair 1 is not a replay of the single-pair run (45,961 against 40,761 rounds), because pair seeds are derived per pair, so the comparison is between samples. The wire bits are 4.09 and 2.73 times those of the single-pair runs, around the factor three that three pairs imply. No pair is broken, and the highest final agreement on one pair is 15.9%, on the slowest pair of R5; we draw no causal conclusion from that. Because every per-pair success is observed as zero, these runs cannot test the product law p^P. They show the latency and bandwidth side of the ensemble and not its security benefit.

Quantity	R5 (B = 1)	R6 (B = 10)
Pair finish rounds	45,961 / 78,290 / 42,478	7,261 / 4,238 / 3,929
Serial rounds (sum)	166,729	15,428
Parallel rounds (slowest pair)	78,290	7,261
Parallel over serial	0.47	0.47
Parallel over single-pair run	1.92	1.28
Wire bits	333,458	308,560
Wire bits over single-pair run	4.09	2.73
Attacker per pair (%)	2.3 / 15.9 / 2.9	2.3 / 3.3 / 3.1

Table 5. Ensemble runs with P = 3. Serial is the sum of the three finish times. Single is the matching single-pair run (R1 for R5, R2 for R6). Attacker is the final best agreement on each pair.

5.6 Reconciliation and key derivation (R1 to R8)

With modulus 4 and L = 57, the 3000 weights of a single pair leak 6,000 bits (2 bits per weight, 4.1% to 7.4% of the tau wire bits) and leave 14,537 bits, and the 9000 weights of three pairs leak 17,999 bits (5.4% and 5.8%) and leave 43,611 bits. The margins over the 256-bit key are 56.8 and 170.4 times under the uniformity assumption of Section 3.6, which is an upper bound. Three runs end with fewer than 100.00% equal weights before reconciliation: R4 (99.97%), R7 (99.90%) and R8 (99.97%), that is 1, 3 and 1 differing weights of 3000. Reconciliation repaired all three, consistent with differences of one, below the repair radius $m/2$ = 2. The leakage of the tau transcript is not counted.